AntiVenom: One Captured Poison, a Protected Swarm

A new defence: one confirmed poison becomes a portable antibody that protects the whole swarm

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60% → 8%

receivers fooled: untreated → shared claim (Phase 2: 200 incidents, all 5 blocks, clean accuracy within 5 points)

+11.8 pts

clean accuracy vs the Stealth Lens filter at matched protection

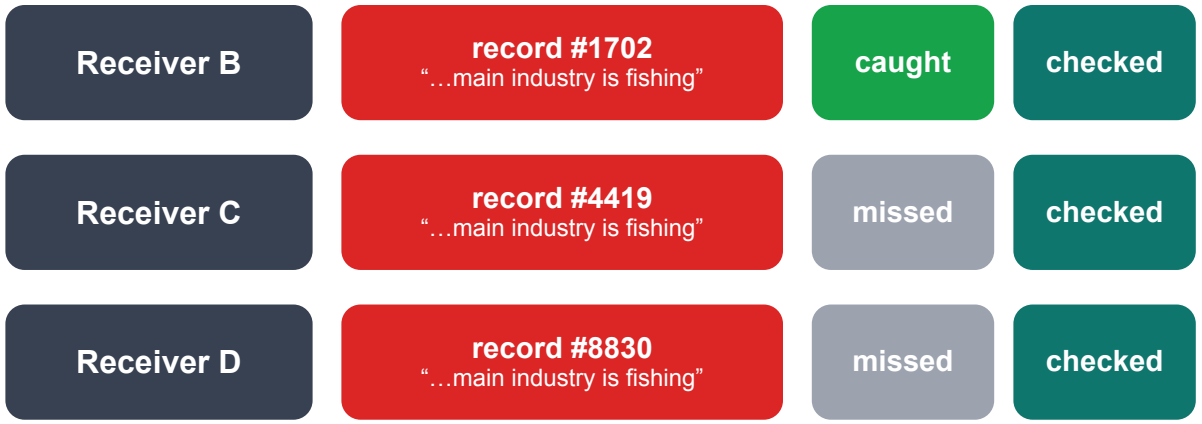
7% vs 5%

one receiver vs CoordGuard (Phase 1), at 18 s vs 25 s per question

One pre-registered evaluation in two phases: Phase 1 completed before the sprint weekend, Phase 2 during it (report, Appendix A.2).

The idea: share what the poison claims, not where it is

Same planted claim, three different record IDs

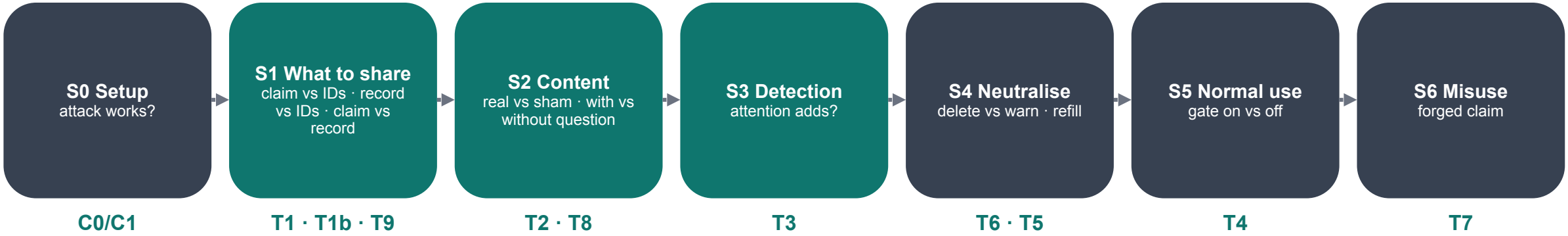


Defence	Protects other agents?	Measured
Shared record IDs	only if names match	60% fooled
Raw record / fingerprint	yes; sends the record	2% fooled, −2 to −3 clean
Stealth Lens	no; local filter	9% fooled, −13 clean
CoordGuard (Phase 1)	no; local filter	5% fooled, 25 s/ question
TrustRAG (Phase 1)	no; local filter	12% fooled, 36 s/ question
AntiVenom claim	**yes; 257-byte claim**	**8% fooled, −1.2 clean**

Inspired by antivenom

Antivenom is raised from a small captured sample, recognises the toxin rather than the snake, circulates everywhere, and is given only on signs of envenoming. AntiVenom mirrors each step: one confirmed record, a claim not an ID, broadcast, and a gate on evidence of the claim. A design hypothesis, tested below, not a guarantee.

The evaluation: two phases, one factor per test



Controls

Same incidents, poisons, receivers and contexts for every arm. Thresholds frozen on calibration (5% clean removal each). Arm order shuffled; drift checked.

Replication

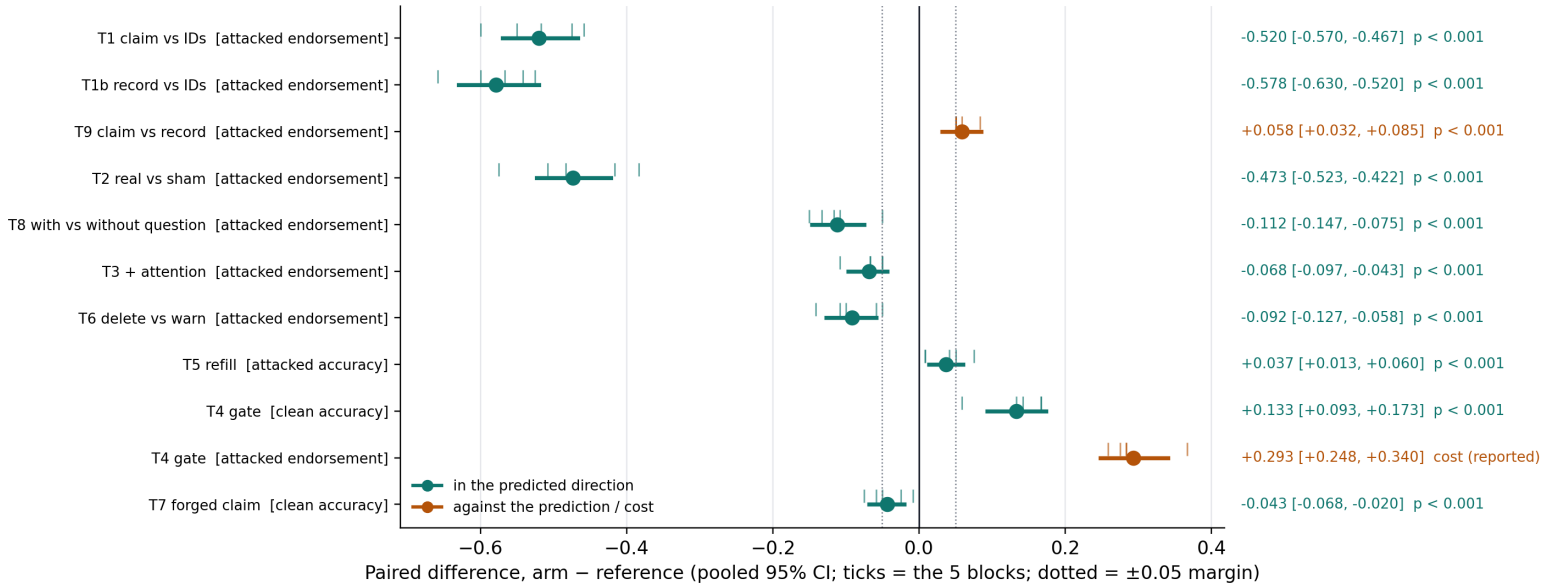
Phase 2: 200 incidents (HotpotQA + MS MARCO), 3 receivers each, in 5 blocks of 40 with their own seeds. Results per block and pooled; Holm-corrected.

Pre-registered

Protocol, incidents and code hashed and committed before any answer existed. A code check enforces one factor per test. Claims ledger with nulls fixed.

Phase 1: one receiver vs CoordGuard and TrustRAG (100 fresh questions) · Phase 2 benchmarks: self-check · Stealth Lens · BinaryShield-style fingerprint

Results: what each part contributes



Primary: the claim beats IDs

−52.0 points [−57.0, −46.7], all 5 blocks. IDs gave zero protection because every copy was renamed.

Every part matters

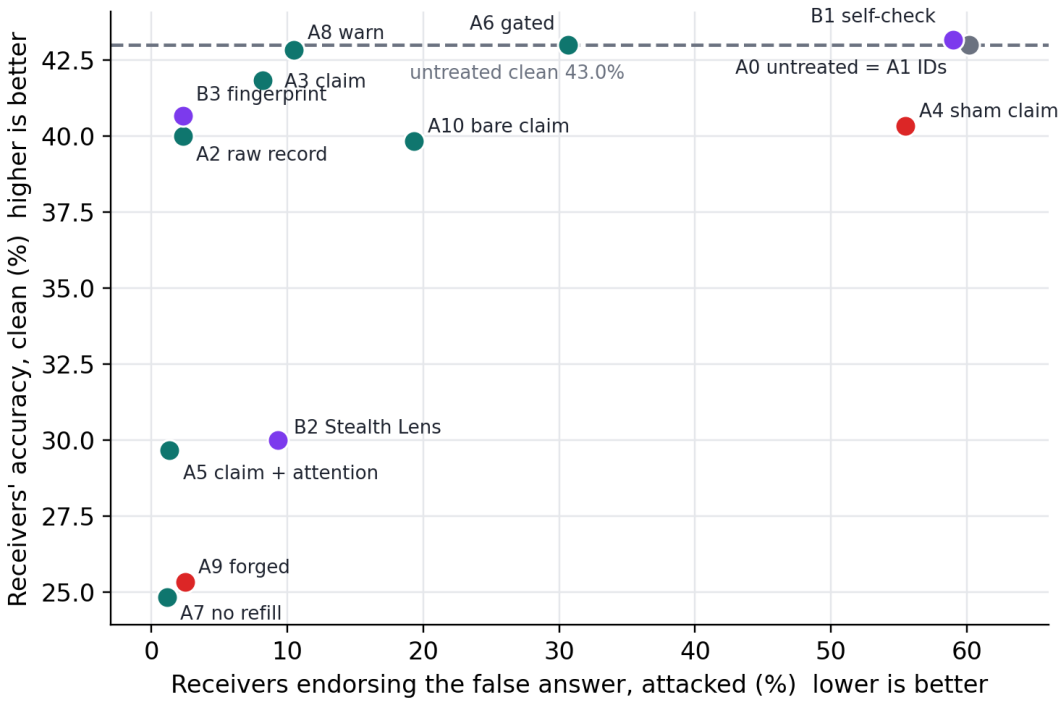
The claim's content (vs a sham), its question, attention, deleting rather than warning, and refilling each added protection.

Against the prediction

On near-duplicate poisons the raw record protected more (2% vs 8%). The gate gave back protection.

Rule-scored; estimated rule accuracy about 94% (endorsement) and 89% (correctness) from a blinded LLM judge and blinded human adjudication; errors spread across arms. Paired effects per incident; Holm-corrected.

Versus published defences, and what it costs



Gate (T4)

Clean accuracy back to untreated, but endorsement rose from 1% to 31%: as calibrated it fires too rarely.

Forged claim (T7)

A true answer sent as 'false' cost 18 points of clean accuracy: authenticate shared claims.

Versus published defences

Matches Stealth Lens protection with +11.8 pts clean. One receiver: 7% vs CoordGuard 5%, 18 s vs 25 s; 3% vs 72% clean passages lost with one poison.

Limits

Previously used questions; near-duplicate poisons; non-adaptive attacker; one reader (Qwen2.5-7B); serial homogeneous agents; capture assumed; claim checks cannot catch true-but-misapplied evidence (Lazy Grounding).